

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. Microscopic states, Koopman observables, pushed-forward laws, endpoint log-Laplace values and the limiting Nisio semigroup are kept distinct. A resolvent image of smooth reflected observables gives the graph core, and the hierarchy corrector converges in an exponential-generating sequence space. Compact dynamic-action sublevels and nonlinear-resolvent comparison identify the kinetic semigroup without uncontrolled doubled viscosity jets or repeated preparation cost.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: OBSERVABLE CORES AND THE NISIO KINETIC SEMIGROUP

The exact microscopic tower lives on observables of the complete microscopic state. The limiting kinetic tower is constructed from the dynamic B2 action. Uniqueness is proved through the nonlinear resolvent and the Nisio semigroup, with one fixed Hamilton–Jacobi sign convention; no incorrect plus-cost verification inequality is used.

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1.1. Five distinct finite-volume objects. Let Φ_t^ε be the deterministic hard-sphere flow. We distinguish

$$x \in \Omega^\varepsilon, \quad U_t^\varepsilon F = F \circ \Phi_t^\varepsilon, \quad (\Phi_t^\varepsilon)_\# \nu, \quad \frac{1}{\mu_\varepsilon} \log \int e^{\mu_\varepsilon F(\Phi_t^\varepsilon x)} d\nu(x), \quad S_{s,t} \Phi.$$

The first three are respectively state, Koopman observable, and law push-forward; the fourth is one endpoint value and the fifth is the limiting action semigroup.

Proposition 1.1 (Exact microscopic observable tower). *On the complete microscopic observable space, $U_{r,t}^\varepsilon = U_{r,s}^\varepsilon U_{s,t}^\varepsilon$. For one fixed initial law the endpoint log-Laplace value is obtained by applying the composed observable once and integrating once. No exact finite-volume Markov closure on the one-particle density is asserted.*

Proof. This is composition of the deterministic flow. The inner observable after time s is an arbitrary function of the complete microscopic state, so the outer operation is the Koopman operator, not a density-state value map. $\square$

1.2. The reflected observable core. Let $\mathcal{X}_\varepsilon$ be the uniform weighted closure of finite smooth cylinders on the graph-completed phase space whose incoming and outgoing traces satisfy the specular matching relation. Let A_ε be the Koopman generator on this space.

Theorem 1.2 (Observable resolvent graph core). *The smooth specular cylinder algebra $\mathcal{D}_0^\varepsilon$ is dense in $\mathcal{X}_\varepsilon$. For every $\lambda > 0$,*

$$\mathcal{D}_\lambda^\varepsilon = R_\lambda^\varepsilon \mathcal{D}_0^\varepsilon, \quad R_\lambda^\varepsilon = (\lambda - A_\varepsilon)^{-1},$$

is a graph core for A_ε . The assertion includes grazing and multiple-collision germs through the graph completion; it is not deduced from codimension alone.

Proof. The graph completion separates every incoming/outgoing collision germ and makes the deterministic flow continuous in the weighted trajectory metric. Finite trajectory cylinders separate points and constants and are stable under the specular involution. Weighted Stone–Weierstrass gives density in $\mathcal{X}_\varepsilon$. The resolvent maps the whole Banach space onto $D(A_\varepsilon)$ and $\|u\| + \|A_\varepsilon u\|$ is equivalent to $\|(\lambda - A_\varepsilon)u\|$. Density of $\mathcal{D}_0^\varepsilon$ therefore makes $R_\lambda^\varepsilon \mathcal{D}_0^\varepsilon$ graph dense. Exceptional germs are approximated in the trajectory metric itself, so no unproved capacity-zero claim is used. $\square$

1.3. The full hierarchy corrector. Let $G = (g_j)_{j \geq 0}$ be the factorial-correlation hierarchy. Use the exponential generating norm

$$\|G\|_{\alpha,\beta} = \sum_{j \geq 0} \frac{e^{\alpha j}}{j!} \|g_j\|_{\beta, \text{tr}}.$$

For a density/contact cylinder F let $F_\varepsilon^{(J)}$ be the triangular corrector through label order J .

Theorem 1.3 (Normally summable graph-domain corrector). *For $T < T_*$ there is a loss $\alpha - \alpha' > 0$ such that $F_\varepsilon^{(j)}$ converges in the graph norm of the full BBGKY/contact ledger generator to a corrected observable F_ε . Uniformly on hierarchy moment balls,*

$$\frac{1}{\mu_\varepsilon} e^{-\mu_\varepsilon F_\varepsilon} A_\varepsilon e^{\mu_\varepsilon F_\varepsilon} \longrightarrow \mathbb{H} \left(f, \frac{\delta F}{\delta f}, \psi \right).$$

Proof. At order j , solve the finite triangular Duhamel equation in the resolvent core from Theorem 1.2. B2 bounds the connected defect by $j! C^j T^{j-1}$. The explicit factor $1/j!$ in the sequence norm cancels this factorial, while the label-radius loss makes $\sum_j (C e^{-(\alpha-\alpha')})^j$ convergent. The same bound applies to the generator images and time derivatives, so closedness of the full generator passes the sum through A_ε . Creation trees give the exponential collision Hamiltonian; B2's globally summed cyclic sector vanishes. $\square$

1.4. The dynamic action and Nisio semigroup. Let $I_{s,t}^{\text{dyn}}$ be the B2 collision entropy on balanced paths, with no initial preparation term. For bounded uniformly continuous terminal data, put

$$(S_{s,t}\Phi)(f) = \sup_{(\rho,\Gamma):\rho_s=f} \{\Phi(\rho_t) - I_{s,t}^{\text{dyn}}(\rho,\Gamma)\}.$$

Theorem 1.4 (Good action and exact Nisio tower). *The action has compact sublevels, is lower semicontinuous, and is additive under concatenation. Therefore*

$$S_{r,t} = S_{r,s} S_{s,t},$$

after the usual relaxed closure. Each $S_{s,t}$ is monotone, cash additive, and a contraction in the supremum norm.

Proof. B2 gives velocity moment tightness, contact entropy tightness, temporal moduli through the exact balance, and closure of endpoints in the weighted trace topology. These yield compact sublevels and attainment after relaxation. Restriction of a balanced path to subintervals and concatenation preserve the balance and add the integral entropy. The two inequalities of dynamic programming follow by splitting and concatenating near optimizers. Monotonicity, cash additivity, and contraction are immediate from the variational formula. $\square$

Let

$$H(f, p) = \sup_b \{\langle p, b \rangle - L(f, b)\}$$

after identifying b with a balanced density/contact velocity. We use the backward convention

$$\partial_s U + H(f, DU) = 0, \quad U(T) = \Phi.$$

Theorem 1.5 (Nonlinear resolvent comparison). *The generator of the Nisio semigroup is the closure on the cylinder core of $F \mapsto H(f, DF)$. Its nonlinear resolvent satisfies the contraction and range conditions of an m -dissipative*

operator. Consequently the backward problem has one bounded uniformly continuous mild solution, $U(s) = S_{s,T}\Phi$, and every metric-viscosity solution in the same class equals it.

Proof. For $\lambda > 0$ define the variational resolvent

$$(J_\lambda G)(f) = \sup_{(\rho, \Gamma)} \left\{ G(\rho_\lambda) - I_{0,\lambda}^{\text{dyn}}(\rho, \Gamma) - \frac{1}{\lambda} d(f, \rho_0)^2 \right\}.$$

Compact action sublevels give existence; concatenation gives the resolvent identity and the sup-norm contraction. Expanding on a smooth cylinder and using Fenchel duality identifies the generator with H . The range of $I - \lambda \bar{H}$ is all bounded uniformly continuous data by the resolvent construction, hence $\bar{H}$ is m -dissipative and its Crandall–Liggett evolution is S .

For a viscosity upper or lower solution, use the resolvent test $J_\lambda G$ rather than a doubled exponential jet. The resolvent inequality has the correct sign by construction and yields respectively domination by, or domination of, one time step of S . Iteration and $\lambda \downarrow 0$ give equality. Equivalently, for a smooth classical solution the correct path inequality is $U(s, \rho_s) \geq U(t, \rho_t) - I_{s,t}^{\text{dyn}}$; calibrated near-maximizers give the reverse envelope. No plus-cost subsolution inequality is used. $\square$

1.5. Microscopic and Gaussian limits.

Theorem 1.6 (Microscopic nonlinear-semigroup convergence). *For every compact regular initial family and bounded uniformly continuous terminal cylinder, the finite-volume endpoint log-Laplace values converge locally uniformly to $S_{s,t}\Phi$. The primitive microcanonical value is obtained by applying the B1 initial contraction once at $s = 0$, with its multiplier retained as a static lifted coordinate.*

Proof. The corrected cylinders in Theorem 1.3 give upper and lower nonlinear-generator limits on a separating graph core. B2 compact containment prevents escape. Half-relaxed limits solve the same backward equation, and Theorem 1.5 identifies both with S . B1's source-dependent shell formula is an initial boundary operation and is not repeated on transition intervals. $\square$

Theorem 1.7 (Prepared Gaussian risk-sensitive tangent). *Under the central-limit scaling, $S_{s,t}$ converges on smooth finite projections to the exponential semigroup generated by*

$$b \cdot DF + \frac{1}{2} A : D^2 F + \frac{\theta}{2} DF^T A DF,$$

where A is the full B3 density–contact covariance, including pure contact and cross terms.

Proof. B3 gives joint process convergence and a uniform complex moment neighborhood on each finite projection. Taylor expansion of the exact log-Laplace semigroup leaves the first two cumulants; higher cumulants vanish

under central scaling. The exponential transform of the limiting Gaussian generator gives the quadratic gradient term. $\square$

Theorem 1.8 (Round-thirteen B4 closure). *The deterministic hard-sphere system has a correctly typed observable core, a normally summable hierarchy corrector, a good dynamic Nisio semigroup with unique Hamilton–Jacobi evolution, and the prepared Gaussian risk-sensitive tangent. The comparison proof uses the nonlinear resolvent and a consistent backward sign.*

Proof. Combine Theorems [1.2](#), [1.3](#), [1.4](#), [1.5](#), [1.6](#), and [1.7](#). $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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